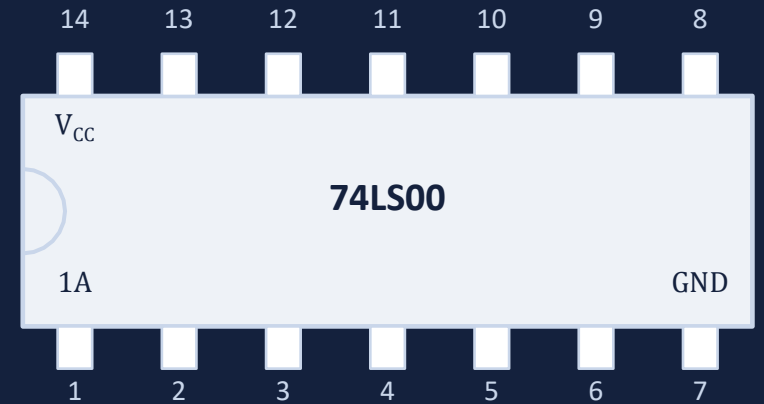


Lecture 5

Integrated Circuits and Logic Families

Levels of integration · data sheets · fan-out, noise margin, power and delay · TTL, CMOS and ECL · CAD and HDLs

Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Chapter 2, §2.9



Today's roadmap

1

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Integrated circuits

SSI to VLSI; data sheets

2

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Family parameters

Fan-out, noise, power, delay

3

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TTL, CMOS, ECL

How the families differ

4

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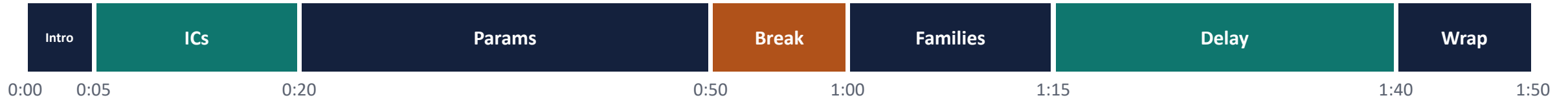
Timing and CAD

Critical path; HDLs

By the end of class you can...

- Read a pin diagram and data sheet
- Compute fan-out and noise margins
- Compare TTL, CMOS and ECL
- Find a circuit's critical path and maximum frequency
- Explain where HDLs fit in modern design

Agenda for today's session



Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome, roadmap	Homework 2 collected
0:05 – 0:20	Integration levels, packages, data sheets	Example 1
0:20 – 0:50	Fan-out, noise margin, power	Examples 2, 3 · Your turn 1
0:50 – 1:00	Break	
1:00 – 1:15	TTL, CMOS and ECL; the CMOS inverter	—
1:15 – 1:40	Propagation delay and the critical path	Example 4 · Your turn 2
1:40 – 1:50	CAD and HDLs; wrap-up	Lab A preview

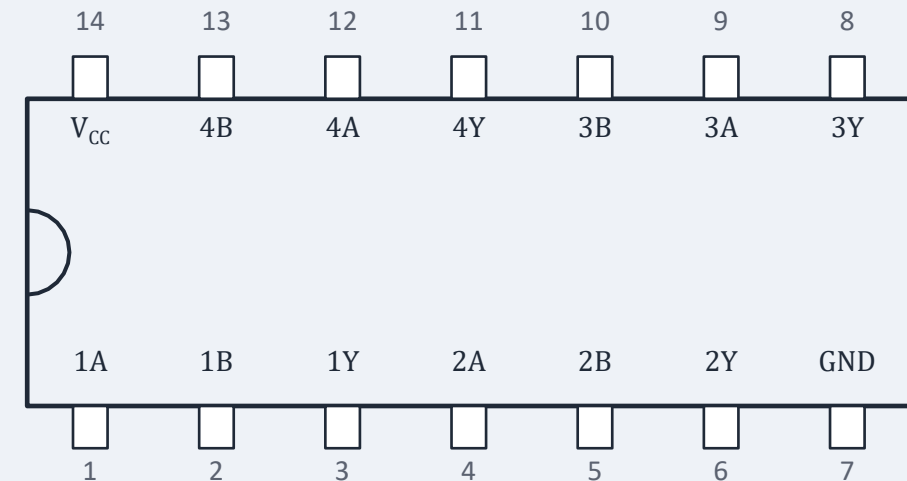
Integrated circuits

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Level	Gates per chip	Examples
SSI	fewer than 10	74LS00 quad NAND, 74LS04 hex inverter
MSI	10 – 1000	decoders, multiplexers, adders, counters
LSI	thousands	small memories, simple processors
VLSI	millions	microprocessors, FPGAs, large memories

A chip (die) is mounted in a package whose pins connect to the gates inside. The part number, e.g. 74LS00, names the function and the family.

74LS00: quad 2-input NAND, top view

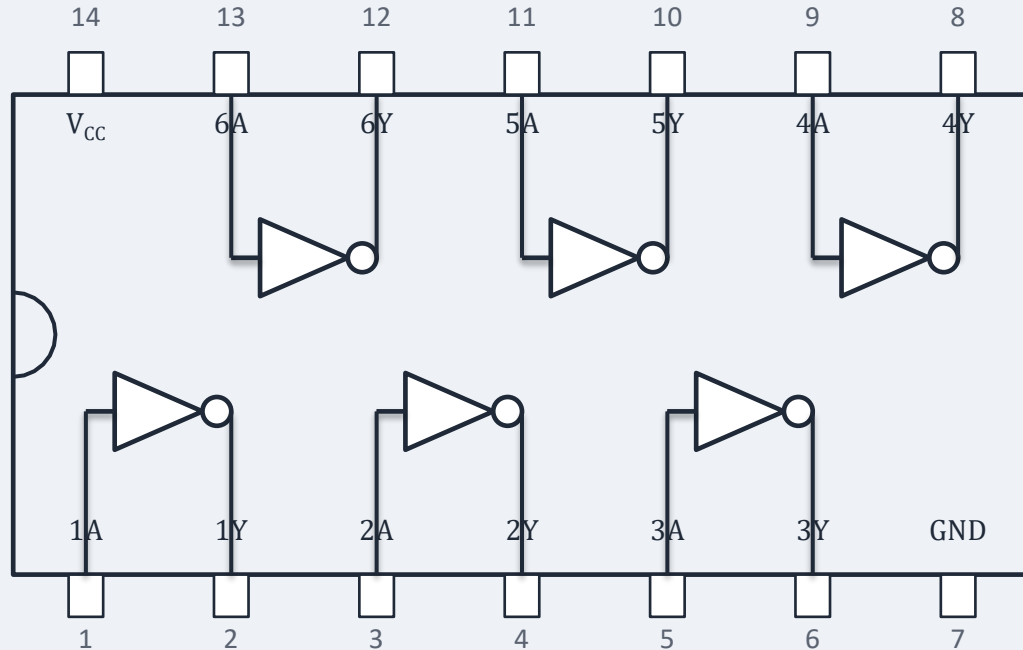


Pin 1 is next to the notch; count counter-clockwise from the top. Find V_{CC} (pin 14) and GND (pin 7) first.

Reading a data sheet: 74LS04

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Six inverters in one 14-pin package



- 1 **Pin diagram:** V_{CC} and GND before anything else.
- 2 **Function table:** note active-low pins (overbar, $\overline{}$).
- 3 **Operating conditions:** supply and temperature limits.
- 4 **Electrical:** V_{OH} , V_{OL} , V_{IH} , V_{IL} and currents.
- 5 **Switching:** $t_{PLH} = 9 \text{ ns}$, $t_{PHL} = 10 \text{ ns}$: use 10 ns.

Example 1 — Counting packages

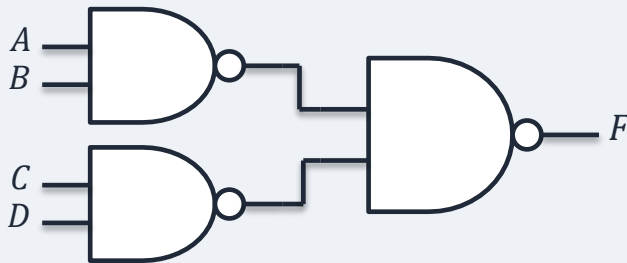
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EXAMPLE

Build $F = AB + CD$ from SSI chips. How many packages does each approach need?

Approach	Chips	Packages
AND and OR	74LS08 + 74LS32	2
NAND only	74LS00	1
Inside an MSI part	already on the board	0 extra

NAND-NAND form of $F = AB + CD$



- 1 Direct form: 2 ANDs (74LS08) and 1 OR (74LS32)
- 2 That is **2 packages**, with 5 of the 8 gates unused
- 3 NAND form: $F = [(AB)'(CD)']'$
- 4 Check with DeMorgan: $(AB)'' + (CD)'' = AB + CD$ ✓
- 5 3 NAND gates fit on one 74LS00 (4 gates): **1 package**
- 6 Fewest literals is not always fewest packages

Parameters of a logic family

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Fan-out

Standard loads one output can drive and still meet its voltage specs.

Fan-in

Inputs on one gate. Large fan-in is slow, so wide functions become trees.

Power

Power drawn per gate (mW). CMOS: nearly zero at rest, rises with frequency.

Delay t_{pd}

Input change to output change (ns). Limits the clock frequency.

Noise margin

Spurious voltage an input tolerates without a wrong output.

Speed × power

Delay × power (pJ): the figure of merit. Faster usually costs power.

Example 2 — Fan-out

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EXAMPLE

A 74LS00 output: $I_{OL} = 8 \text{ mA}$, $I_{OH} = 0.4 \text{ mA}$. Each 74LS input: $I_{IL} = 0.4 \text{ mA}$, $I_{IH} = 20 \text{ } \mu\text{A}$. Fan-out?

Why two states?

LOW output: the gate SINKS the current of every load.

HIGH output: the gate SOURCES current into every load.

It must work in both, so take the smaller count.

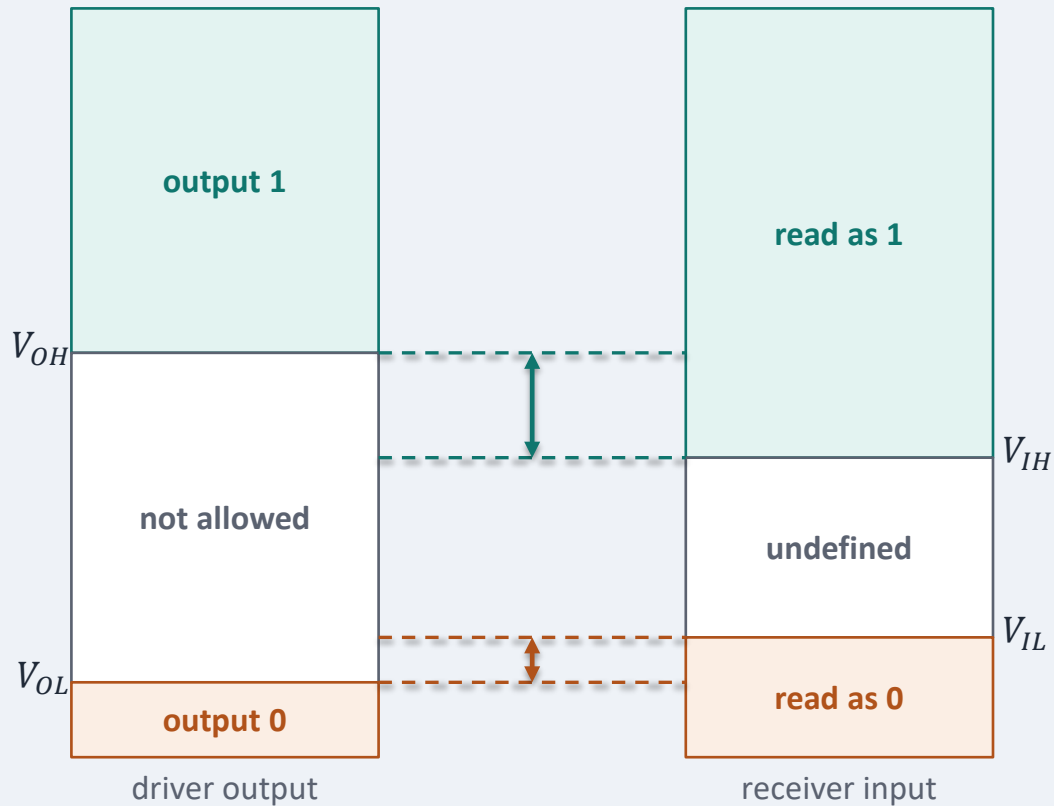
$$N = \min(I_{OL}/I_{IL}, I_{OH}/I_{IH})$$

- 1 Use $N = \min(I_{OL}/I_{IL}, I_{OH}/I_{IH})$
- 2 LOW: $8 \text{ mA}/0.4 \text{ mA} = 20 \text{ loads}$
- 3 HIGH: same units first: $0.4 \text{ mA} = 400 \text{ } \mu\text{A}$
- 4 $400 \text{ } \mu\text{A}/20 \text{ } \mu\text{A} = 20 \text{ loads}$
- 5 Fan-out = $\min(20, 20) = 20$
- 6 Overloaded LOW: V_{OL} rises and the margin vanishes
- 7 Need more? Add a buffer or split the load

Noise margin

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Guaranteed outputs versus required inputs



Definitions

$$NM_H = V_{OH} - V_{IH}$$

$$NM_L = V_{IL} - V_{OL}$$

The gap between what a driver guarantees and what a receiver needs.

The family's margin is the smaller of the two.

Example 3 — Noise margins

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EXAMPLE

74LS: $V_{OH} = 2.7$, $V_{OL} = 0.5$, $V_{IH} = 2.0$, $V_{IL} = 0.8$ V. Find the noise margins and compare 5 V CMOS.

	74LS	5 V CMOS	
V_{OH}	2.7	4.9	V
V_{IH}	2.0	3.5	V
V_{IL}	0.8	1.5	V
V_{OL}	0.5	0.1	V
NM_H / NM_L	0.7 / 0.3	1.4 / 1.4	V

- 1 $NM_H = V_{OH} - V_{IH} = 2.7 - 2.0 = \mathbf{0.7\text{ V}}$
- 2 $NM_L = V_{IL} - V_{OL} = 0.8 - 0.5 = \mathbf{0.3\text{ V}}$
- 3 The margin is the smaller: **0.3 V** (LOW is weak)
- 4 CMOS: $NM_H = 4.9 - 3.5 = 1.4\text{ V}$
- 5 CMOS: $NM_L = 1.5 - 0.1 = 1.4\text{ V}$
- 6 Compare: $1.4/0.3 \approx 4.7 \times$ the margin
- 7 and the same in both states

Your turn 1 — Loading and margins

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YOUR TURN

≈ 10 min

- (a) 74HC at 4.5 V: $V_{OH} = 4.4$, $V_{OL} = 0.33$, $V_{IH} = 3.15$, $V_{IL} = 1.35$ V. Find NM_H and NM_L .
- (b) Standard TTL: $I_{OL} = 16$ mA, $I_{OH} = 0.4$ mA, $I_{IL} = 1.6$ mA, $I_{IH} = 40$ μ A. Find the fan-out.
- (c) A 74LS04 draws $I_{CC} = 6.6$ mA at 5 V. Find the power per package and per gate.

State which state (HIGH or LOW) limits each answer.

Your turn 1 — Solution

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SOLUTION

- 1 (a) $NM_H = V_{OH} - V_{IH} = 4.4 - 3.15 = \mathbf{1.25\text{ V}}$
- 2 $NM_L = V_{IL} - V_{OL} = 1.35 - 0.33 = \mathbf{1.02\text{ V}}$
- 3 The smaller is 1.02 V: the LOW state limits 74HC
- 4 (b) LOW: $I_{OL}/I_{IL} = 16/1.6 = 10$
- 5 HIGH: $0.4\text{ mA} = 400\text{ }\mu\text{A}$; $400/40 = 10$
- 6 Fan-out = $\min(10, 10) = \mathbf{10}$
- 7 (c) $P = V_{CC} I_{CC}$
- 8 $= 5\text{ V} \times 6.6\text{ mA} = \mathbf{33\text{ mW}}$ per package
- 9 A 74LS04 holds 6 inverters
- 10 $33/6 = \mathbf{5.5\text{ mW}}$ per gate

Break

10 minutes — we resume at the 1:00 mark

Next: TTL, CMOS and ECL, and propagation delay

TTL, CMOS and ECL compared

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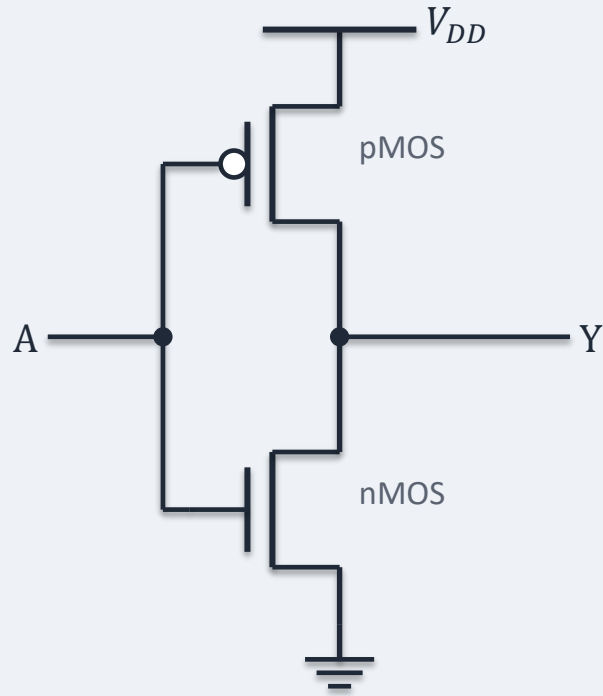
	TTL (74LS)	CMOS (74HC, 4000)	ECL
Transistors	bipolar, saturating	MOSFET pairs	bipolar, never saturate
Supply	5 V	1.8–15 V	–5.2 V
Speed	about 10 ns	a few ns (modern)	fastest, < 1 ns
Power	about 2 mW / gate	≈ 0 static, CV^2f	highest
Noise margin	0.3 V	$\approx 30\text{--}40\%$ of V_{DD}	small
Open input	floats HIGH	undefined: never leave	—
Where used	lab, legacy boards	every VLSI chip, FPGAs	very high speed

Tie every unused input to V_{CC} or ground. TTL outputs are totem-pole: never wire two outputs together.

The CMOS inverter

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A pMOS and an nMOS in series



A	pMOS	nMOS	Y
0	on	off	1
1	off	on	0

Why CMOS wins

One transistor is always off, so no current flows from V_{DD} to ground at rest.

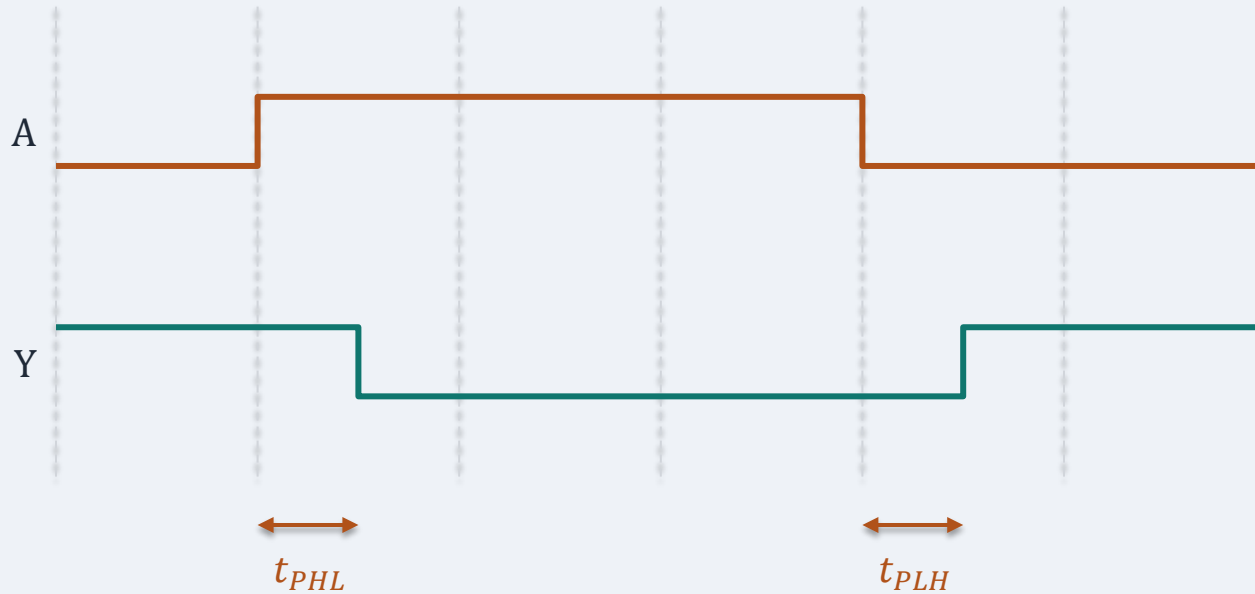
Power is spent charging the load: $P = CV_{DD}^2 f$.

NAND and NOR need 4 transistors; AND and OR need 6. That is why NAND and NOR are the natural gates.

Propagation delay

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Input and output of an inverter



Measured between the 50 % points of the input and output edges.

Two delays

t_{PHL} : output falls. t_{PLH} : output rises.

Use the larger for worst-case timing.

Critical path

The slowest input-to-output path sets the fastest the inputs may change:

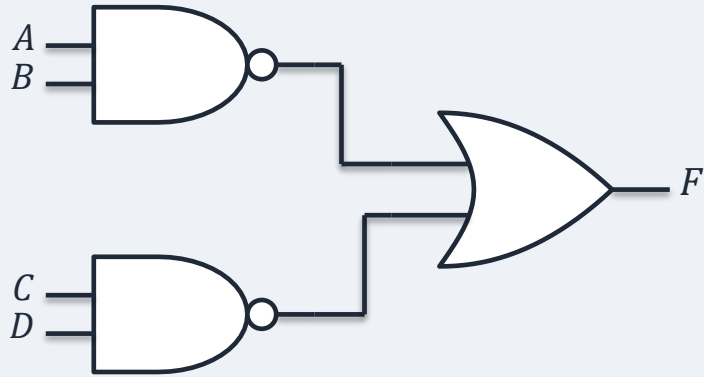
$$f_{max} = 1/t_{critical}$$

Example 4 — Critical path

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EXAMPLE

$F = (AB)' + (CD)'$: two NANDs (10 ns) feed an OR (12 ns). Find the worst delay and f_{max} .



Every path: NAND → OR.

Add an inverter on A only and the A path becomes NOT → NAND → OR.

- 1 Each input passes one NAND, then the OR.
- 2 Each path: $10 + 12 = 22$ ns; all four are equal
- 3 $f_{max} = 1/22$ ns = **45.5 MHz**
- 4 Add a 10 ns inverter on A: $10 + 10 + 12 = 32$ ns
- 5 The longest path, 32 ns, is now critical
- 6 $f_{max} = 1/32$ ns = **31.25 MHz**
- 7 Unequal paths also risk a glitch on F

Your turn 2 — Delay and power

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YOUR TURN

≈ 10 min

- (a) Signal A passes an inverter, then two NAND gates in series; B passes only the two NANDs. Every gate has $t_{pd} = 10$ ns. Find the critical path and f_{max} .
- (b) A CMOS node has $C = 15$ pF, $V_{DD} = 3.3$ V and switches at $f = 50$ MHz. Find its dynamic power.

Dynamic power: $P = CV_{DD}^2 f$.

Your turn 2 — Solution

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SOLUTION

- 1 (a) Path A: NOT + NAND + NAND = $10 + 10 + 10 = 30$ ns
- 2 Path B: NAND + NAND = $10 + 10 = 20$ ns
- 3 Critical path = the longest: A, **30 ns**
- 4 $f_{max} = 1/(30 \times 10^{-9}) = \mathbf{33.3\text{ MHz}}$
- 5 (b) Write the formula: $P = CV_{DD}^2 f$
- 6 Square the supply: $3.3^2 = 10.89\text{ V}^2$
- 7 Substitute: $15 \times 10^{-12} \times 10.89 \times 50 \times 10^6$
- 8 $= 8.17 \times 10^{-3}\text{ W} = \mathbf{8.17\text{ mW}}$
- 9 Halve V_{DD} and power falls by 4: the reason chips run at low voltage

Computer-aided design and HDLs

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Implementation targets

Full-custom IC: every transistor placed by hand

ASIC: standard cells, fixed after fabrication

FPGA: reprogrammable logic blocks (your Nexys board)

PLD: small programmable arrays (Lecture 27)

Designs with millions of gates are written in an HDL, simulated, then synthesized by CAD tools.

Verilog: the circuit of Example 1

```
module ex1 (input A, B, C, D,  
            output F);  
    assign F = (A & B) | (C & D);  
endmodule
```

Verilog and VHDL are IEEE standards.

An HDL describes hardware; it is not a program that runs step by step. Lab B starts using it.

Concept check

1 Why is fan-out the smaller of the LOW and HIGH counts?

2 Which 74LS state has the smaller noise margin?

3 Why does a CMOS gate draw almost no power when idle?

4 Why must unused CMOS inputs never be left open?

Key takeaways

Fan-out

$$\min(I_{OL}/I_{IL}, I_{OH}/I_{IH})$$

Noise margin

$$NM_H = V_{OH} - V_{IH}$$

Noise margin

$$NM_L = V_{IL} - V_{OL}$$

CMOS power

$$P = CV_{DD}^2 f$$

Max frequency

$$f_{max} = 1/t_{critical}$$

Families

ECL fast · CMOS dense

Real chips have limits: read the data sheet before you wire.

Homework #3 due at the start of Lecture 7

Next: Karnaugh maps: two, three and four variables